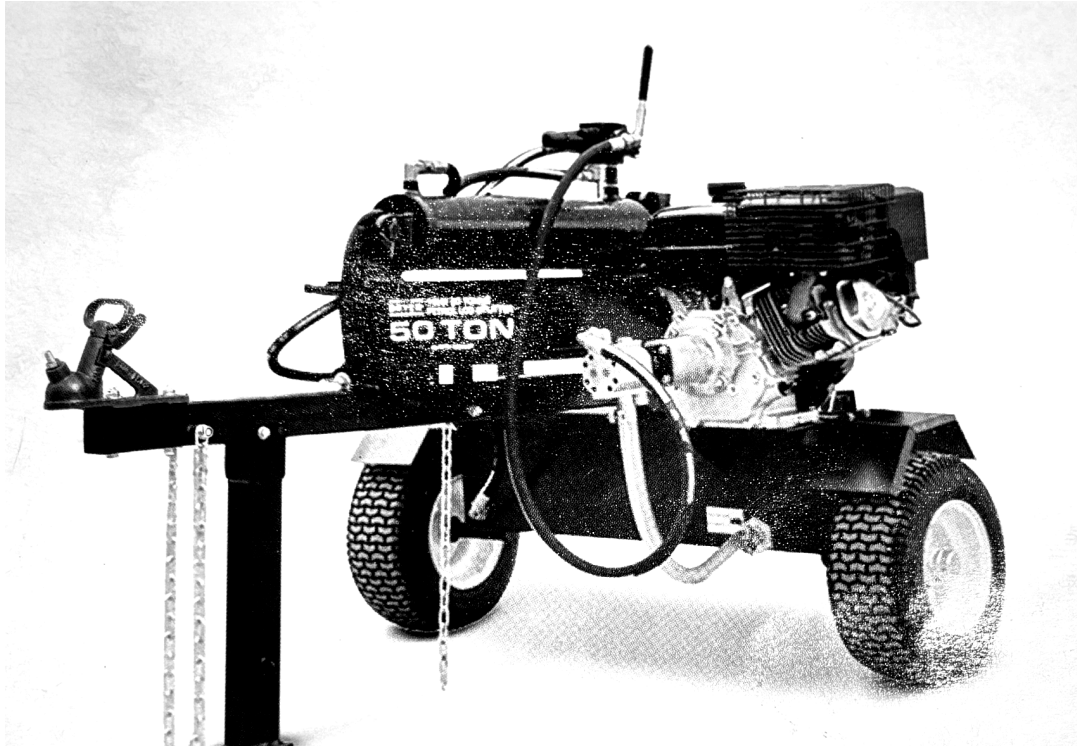


15HP 50 Tonne Petrol Log Splitter

Model **BS50TON**



IMPORTANT

Please make certain that the person who is to use this equipment carefully reads and understands these instructions before starting operations.

Contents

- 1** Important safety information
 - 2** Symbols
 - 3** Technical specifications
 - 4** Before use — inspection & maintenance
 - 5** Work site & log splitter set-up
 - 6** Splitting operation
 - 7** Storage
 - 8** Troubleshooting
 - 9** Assembly
 - 10** Exploded parts diagram
-

FLUIDS AT A GLANCE

Petrol	Unleaded 92#
Engine oil	SAE 10W-30
Hydraulic oil	10 wt AW32, ASLE H-150, or ISO 32 oil — 20 L

SECTION 1

Important safety information

WARNING — HITCH COUPLER

This trailer's Hitch Coupler **must** be properly secured to the hitch ball of the towing vehicle. After assembly and attachment, pull up and down on the Hitch Coupler to make sure the hitch ball is fitting snugly in the Hitch Coupler. **There must be no play between the hitch ball and Hitch Coupler.** If there is play, tighten the Adjustment Nut until no play is present. If the Adjustment Nut is too tight, the Handle will not lock. **Carefully read and follow the complete instructions in this manual BEFORE setup or use.**

If the Coupler is not secured properly, the ball could come loose while the trailer is in motion, possibly causing property damage, SERIOUS PERSONAL INJURY, or DEATH.

Fuel and oil

Gasoline engine	Unleaded 92#
Gasoline engine oil	SAE 10W-30
Hydraulic oil	10 wt AW32, ASLE H-150, or ISO 32 oil — 20 L

Unpacking

 **WARNING:** To avoid injury from unexpected starting or electrical shock during unpacking and setting up, do not plug the power cord into a source of power. This cord must remain unplugged whenever you are working on the machine.

Separate all parts from packing materials and check each one with the illustration and the list of Loose Parts to make certain all items are accounted for before discarding any packing material.

If any parts are missing, do not attempt to assemble the machine, plug in the power cord or turn the switch on until the missing parts are obtained and are installed correctly.

SECTION 2

Symbols



Read the manual



Double insulation



Wear ear protection



Wear eye protection



Wear dust mask



Do not expose to rain or use in damp locations

SECTION 3

Technical specifications

Engine	15 HP, 4 stroke, OHV, air cooled
Start	Manual (recoil) start
Hydraulic cylinder size	5.0"
Oil capacity	20 litres
Gear pump	2 stage, 16 GPM flow
Valve	Automatic return control valve with safety release function
Maximum splitting force	50 ton
Wedge height	8 inch (200 mm)
Split and return cycle time	18 seconds
Maximum log length	24 inch (610 mm)
Working position	Horizontal and vertical
Wheels and tyres	16 inch pneumatic × 6.5 inch
Coupler	50 mm cast steel with safety chains
Maximum recommended towing speed	40 km/h
Package dimensions	210 cm × 54 cm × 54 cm
Package weight	315 kg

Note: Before using the tool, read the instruction book carefully.

SECTION 4

Before use — inspection & maintenance

Step One: Inspect and maintain the log splitter before each use.

If the log splitter has been used previously, it must be inspected and maintained BEFORE EACH SUBSEQUENT USE.

WARNING

ALWAYS shut off the tractor, relieve system pressure, and disconnect hydraulic lines before inspecting, cleaning, adjusting, or repairing the splitter. Relieve system pressure by moving the Split Control Lever back and forth several times.

Important: If a part needs replacement, only use parts that meet the manufacturer's specifications. Replacement parts that do not meet specifications may result in a safety hazard or poor operation of the log splitter.

1. Tractor off / relieve hydraulic pressure	Perform all inspections and repairs with the tractor off and hydraulic system pressure relieved. 1. Turn off tractor engine. 2. Relieve all hydraulic system pressure by moving the Split Control Lever and the tractor hydraulic control valve handle back and forth several times. 3. Disconnect the hydraulic lines.
2. Remove debris	Remove debris from moving parts. Debris on moving parts can cause excess wear. Clear debris from the slide beam, wedge, and endplate.
3. Mechanical parts	Check to be sure all nuts and bolts are tight to make sure the log splitter is in safe working condition.
4. Hydraulic components	1. Visually inspect all hoses and clamps/fittings for cracks, fraying, kinks, or other damage. 2. Check all components for oily residue, which may indicate a leak. Do NOT operate the log splitter if there is any indication of damage or oily residue. Small leaks in hydraulic lines can cause severe injuries and can also be an indication of catastrophic failure in the near future. The life of hydraulic hoses may be from a few months to a few years, depending on use and storage patterns. WARNING: High fluid pressures and temperatures are developed in hydraulic log splitters. Hydraulic fluid escaping through a pin hole sized opening can burn or puncture skin, resulting in wounds that could cause blood poisoning, infection, disability, gangrene, amputation, or death. Therefore, the following instructions should be heeded at all times when inspecting or servicing the hydraulic components of the log splitter: • Stop the engine and move all control valve handles back and forth to relieve pressure before changing or adjusting hydraulic system components such as hoses, tubing, fittings or other components. • NEVER check for leaks with your hand. Leaks can be located by holding a piece of cardboard or wood (at least two feet long) with your hand at one end and passing the other end over the suspected area (wear eye protection). Look for discolouration of the cardboard or wood. • If injured by escaping fluid, no matter how small the wound is, see a doctor at once. A typical injection injury may be a small puncture wound that does not look serious. However, severe infection or reaction can result if proper medical treatment is not administered immediately by a doctor who is familiar with injection injuries.
5. Shields / guards	Replace all guards and shields after servicing the log splitter.

SECTION 5

Work site & log splitter set-up

WARNING

It is important to select an appropriate work site and properly set up the log splitter in order to minimise the risk of slips and falls, equipment rolling or tipping over, carbon monoxide poisoning, and accidental fires.

1. Select location	Select an appropriate location for operating the log splitter. Requirements: 1. Dry, level surface with good footing. Stay clear of areas with mud, ice, tall grass, weeds, or brush. 2. Outdoors, away from air intakes. WARNING: The running engine gives off carbon monoxide, a poisonous gas that can kill you. You CANNOT smell it, see it, or taste it. • ONLY run the log splitter OUTDOORS and away from air intakes. NEVER run the log splitter inside homes, garages, sheds, or other buildings or semi-enclosed spaces. These spaces can trap poisonous gases, EVEN if you run a fan or open windows. • If you start to feel sick, dizzy, or weak while using the log splitter, shut off the engine and get to fresh air RIGHT AWAY. See a doctor — you may have carbon monoxide poisoning.
2. Fire procedures	Take the following precautions against fire: keep a fire extinguisher available (classified for both ordinary combustibles and flammable liquids) as a precautionary measure when operating the log splitter in dry areas.
3. Position 7 ft from any combustibles or flammable liquids	Position the exhaust and other hot elements of the engine at least 7 ft. from combustible objects during splitter operation. Hot exhaust fumes could cause fire.
4. Engage parking brake	Make sure the tractor's parking brake is engaged before operating the log splitter. WARNING: If the parking brake is not engaged, the tractor could roll while you are operating the log splitter. Personal injury could result.

SECTION 6

Splitting operation

WARNING

Before starting the log splitter, review the following instructions and safety information for safe operation. Failure to follow these rules may result in serious injury to the operator or bystanders from moving parts that crush, cut, or entangle; from flying objects, fire, falling or tripping; or from carbon monoxide poisoning.

General safety rules

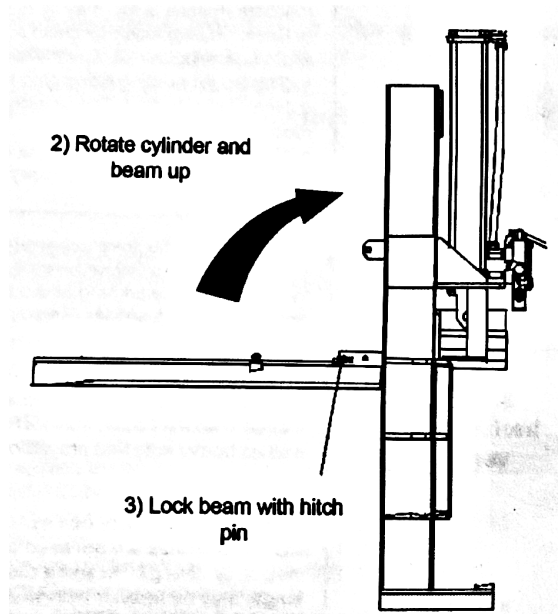
- **Read manual.** Do not allow anyone to operate the log splitter who has not read the Owner's Manual or has not been instructed on its safe use. The owner should instruct all operators in safe log splitter operation.
- **Age restrictions.** NEVER allow anyone under 16 years old to operate the log splitter. Children 16 years and older must be trained and supervised by a trained adult.
- **Intended use.** Log splitters should only be used for splitting wood logs, lengthwise with the grain. Do not use for other purposes as unforeseen hazards may result.
- **Modifications.** NEVER modify or alter the log splitter in any way. Modifications can create serious safety hazards and will void the warranty.
- **Attachments.** NEVER add attachments to the splitter, except for authorised accessories supplied by the manufacturer with instruction for safe installation and use.
- **Remote control.** NEVER attach a rope, cable, or other remote device to the splitting control.
- **Splitting wedge.** NEVER attempt to change the height or speed of the splitting wedge.
- **Safety equipment / controls.** Always operate the log splitter with all safety equipment in place and in good working order, and all controls properly adjusted for safe operation.
- **Know how to stop.** Be thoroughly familiar with all controls and with the proper use of the equipment. Know how to stop the log splitter and relieve system pressures quickly if needed.
- **Operating speed.** Always operate the log splitter at the manufacturer's recommended speed. Adjust the engine speed so the flow rate does not exceed 11 GPM / 16 GPM. Higher flow rates make the beam slide move dangerously fast.
- **Daylight only.** Only use the log splitter in daylight so you can see what you are doing.
- **Smoking / sparks.** NEVER smoke while operating the log splitter, and never operate near sources of sparks or flames.

- **Under the influence.** NEVER operate, or let anyone else operate, the log splitter while under the influence of alcohol, drugs, or medication.
- **Unattended.** NEVER leave the machine unattended while the engine is running.
- **Adjusting / repairing.** Turn the engine off before cleaning, repairing, or adjusting the log splitter. Before changing or adjusting hydraulic system components such as hoses or fittings, first relieve hydraulic line pressure by moving the Split Control Lever and the hydraulic control valve handles back and forth several times, and then disconnect hydraulic lines.
- **Replace labels.** Always make sure safety labels are in place and in good condition.

1. Put on protective clothing / gear	Wear the following protective clothing and safety gear: • Eye protection. Always wear safety glasses or goggles when operating the machine. Pieces of log may fly out and serious eye injury can occur. • Boots. Falling logs can crush feet. Always wear safety shoes or heavy boots when operating or helping to load logs. • Gloves. Wear snug fitting gloves without draw strings or loose cuffs. • Hearing protection. The use of ear plugs or other hearing protection device is recommended. • No loose / dangling apparel. Loose or dangling apparel can become entangled in moving parts. Never wear jewellery or loose-fitting clothing.
2. Engage parking brake	Secure the log splitter from unintended movement — make sure the tractor's parking brake is engaged before operating the log splitter. WARNING: If the parking brake is not engaged, the tractor could roll while you are operating the log splitter. Personal injury could result.
3. Set to horizontal or vertical	Set the log splitter into either the horizontal or vertical splitting position. The HORIZONTAL splitting position is used for lighter logs that can be easily loaded onto the beam. The VERTICAL splitting position is used for light logs as well as heavy logs that are difficult to load onto the beam. Note: Musculoskeletal injury can result from lifting logs onto the log splitter if proper lifting techniques are not used or the logs are too heavy for a person's size, weight, or strength. In some cases, logs as small as 8" in diameter and 14" in length may be heavier than what some persons should be repeatedly lifting onto the splitter. The use of the vertical splitting position can greatly reduce the need to lift logs onto the splitter. Employers are advised to consider NIOSH lifting guidelines when assigning employees to log splitting tasks for an extended period of time. WARNING: NEVER change splitting positions with the engine running. You may contact the muffler and receive serious burns. Make sure the beam is locked securely in the horizontal position with the hitch pin.

3a. Set to vertical position

1. Unlock the beam by removing the hitch pin.
2. Rotate the cylinder and beam until the beam is vertical (NOTE: the beam is heavy).
3. Lock into position with the hitch pin again.



Rotate the cylinder and beam up, then lock the beam with the hitch pin.

3b. Return to horizontal position

1. Unlock the beam by removing the hitch pin.
2. Rotate the cylinder and beam down carefully in a controlled manner until the beam is horizontal.
3. Lock the beam in the horizontal position with the hitch pin.

WARNING: Crush hazard. The beam is heavy — do not let it just drop. It could crush fingers or cause damage to the splitter.

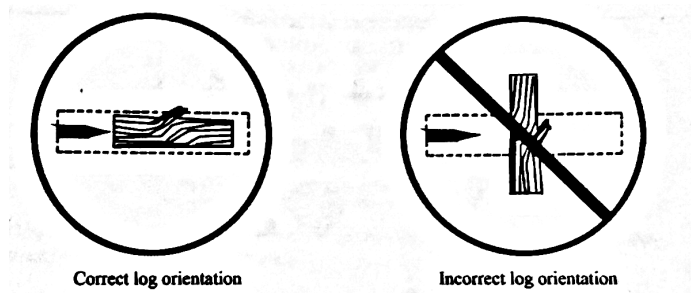
4. Load log

Load the log onto the beam with a cut end against the endplate — positioned for a lengthwise cut.

Notes:

1. The log splitter is designed only for cutting lengthwise with the grain, NOT for cutting across the grain.
2. This log splitter is designed for cutting logs only up to a limited length and diameter. Larger diameter logs can get stuck on the wedge and longer logs will not fit on the beam.

Machine	Max. length	Max. diameter
24 ton	24 inch	16 inch
30 ton	24 inch	18 inch
40 ton	24 inch	20 inch
50 ton	24 inch	22 inch

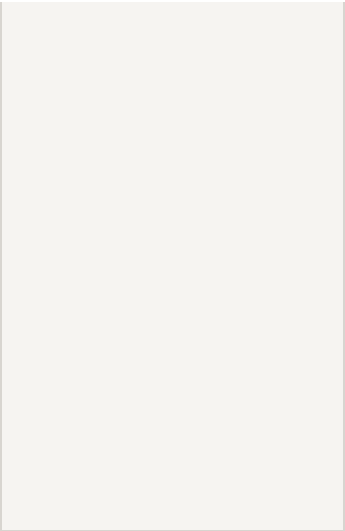


Correct log orientation (left) and incorrect log orientation (right).

WARNING: ALWAYS keep hands and feet away from the endplate, wedge, and partially split logs while loading, operating, and unloading the log splitter.

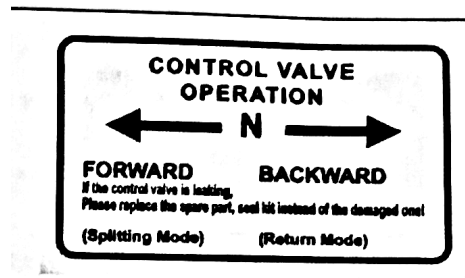
Important safety instructions:

- **Hold bark side.** Hold the bark side of logs when loading or positioning, never the ends. Never place your hands or any part of your body between a log and any part of the log splitter.
- **Note for vertical position loading.** Place the log on the endplate and turn it until it leans against the beam and is stable. If the log is too big or oddly shaped, stabilise the log with wooden shims between the log and the endplate or ground. DO NOT use your leg or knee to stabilise the log. NEVER stabilise the log by placing your hand on top of the log.
- **Wedge moving.** NEVER load or unload logs while the wedge is moving.
- **Straddling / reaching across.** Never straddle, reach across, or step over the beam while the engine is running and the log splitter is in the horizontal position. You could trip, actuate the controls, and get seriously injured.

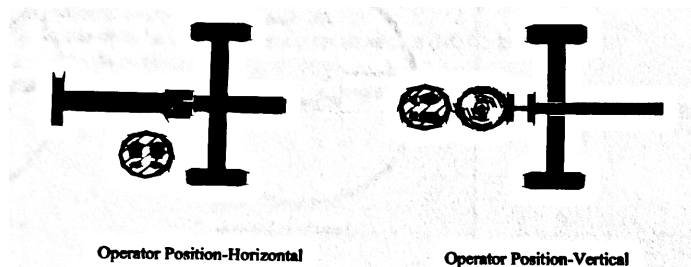
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- **Unsplit log pile.** Do not pile logs to be split in a place that will make you reach across the log splitter in order to load them.
 - **Square log ends.** Logs that are not cut square can slide out while splitting and become a safety hazard or cause excessive force to log splitter components. Use a chainsaw to cut logs square on each end before attempting to split them.
 - **Single log.** Never attempt to split more than one log at a time. Pieces of log can unexpectedly be thrown from the machine causing serious injury.
 - **Split along grain.** Do not use the log splitter to split logs across the grain. Doing so will damage the log splitter and could also cause pieces of log to be thrown, injuring the operator or bystanders.

5. Extend wedge

Move the Split Control Lever toward the endplate to extend the wedge and split the log.



Control valve: forward = splitting mode, backward = return mode, N = neutral.



Operator position — horizontal (left) and vertical (right).

Important safety instructions:

- **Operator position.** ALWAYS operate the log splitter from the manufacturer's indicated OPERATOR POSITION (see diagram above). Other positions are unsafe because they can increase the risk of injury from crushing, cutting, flying objects, or burns.
- **Remove hands.** Remove both hands from the log before activating the Split Control Lever.
- **Hand activate.** Use only your hand to operate the Split Control Lever. Never use any other body part, or a rope, cable, or other remote device to actuate the control.
- **Second person.** Many accidents occur when there is more than one person involved in loading and operating the log splitter. Only one person should operate the controls. If a second person is assisting in loading logs, the operator must NEVER actuate the Split Control Lever until the assistant and all bystanders are at least 10 feet away. NEVER allow an assistant to hold the log in place while the operator actuates the Split Control Lever.

6. Stop wedge

Release the Split Control Lever to stop wedge movement when the log is split.

WARNING: Cracks in logs can close quickly and pinch fingers. Keep fingers away from any cracks that open in partially split logs.

Important: stuck log procedure	If a log does not split completely and becomes stuck on the wedge, follow the instructions below to remove the log. A log can become stuck to the wedge if the wedge becomes embedded in the log and the log doesn't split and separate. This can happen if the log is too stringy or tough to split completely. A stuck log will move back with the wedge on the initial attempt to retract the wedge. If this happens, STOP retracting the wedge immediately and follow the directions below. WARNING: NEVER attempt to remove a stuck log by: • Using the hydraulic force of the splitter • Modifying the splitter • Adding attachments to the splitter Personal injury could result from log or metal pieces flying out at high speed toward the operator or bystanders, or the splitter could become damaged. 1. If there is already 1" or more of clearance between the log and endplate, go to step 2. Otherwise, retract the wedge just enough to remove pressure between the log and endplate about 1" clearance. (Move the split control to return, allow the wedge to move about 1", then stop the wedge by moving the split control to neutral "N".) 2. Turn the engine OFF. 3. Remove the stuck log from the wedge manually with a pry bar or a sledgehammer. Important: Be extremely careful as log pieces may fly off as they separate from the wedge. Make sure bystanders are clear and wear safety goggles. 4. Do not attempt to resplit a stuck log once it has been removed from the wedge. Manually split with an axe, or cut with a chainsaw.
7. Return wedge	Move the Split Control Lever away from the endplate to return the wedge. Once the control valve is actuated in the return direction, the wedge is designed to keep returning by itself completely and then stop automatically. WARNING: Stay clear while the wedge is returning. It is still powerful enough on the return stroke to cause serious injury.
8. Remove split wood	Remove split wood from the area. Move each log away from the log splitter after it is split. Split logs left near the log splitter are a trip hazard.
9. After use	Return the splitter to the horizontal position. If in the vertical position, return the log splitter to the horizontal position for greater stability and to prepare for transportation.

SECTION 7

Storage

Follow the instructions below for storing your log splitter between uses.

1. Retract wedge	Retract the wedge completely to keep the rod protected from corrosion.
2. Wipe with oily rag	Wipe the beam and wedge with an oily rag to prevent corrosion.
3. Splitter storage location	Store the log splitter in a dry location away from corrosive material. Note: Do not store the log splitter near fertiliser or similar corrosive materials.

SECTION 8

Troubleshooting

WARNING

Before troubleshooting or attempting to service, read the following safety instructions to avoid serious injury to the operator or bystanders from moving parts that can crush or cut, cause burns, fire or explosion, or escaping high pressure hydraulic fluid.

Important safety instructions

Engine off. Always make sure the engine is off before cleaning, repairing or adjusting the splitter, except as recommended by the manufacturer.

Hydraulic safety

1. Pour in at least 20 L of hydraulic fluid before using this machine for the first time.
2. We sincerely recommend you choose ESSO "Nuto Hzz" or Royal Dutch Shell "Tellus Tzz" hydraulic fluid for your machine.
3. Before starting the engine, pull the rope of the engine several times to make sure the system is filled with the hydraulic fluid; this may relieve the pressure for the hydraulic parts.
4. After using for the first time, clean the filter, and make sure to do this cleaning regularly in order to avoid blocking the filter.

High fluid pressures and temperatures are developed in hydraulic log splitters. Hydraulic fluid escaping through a pin hole sized opening can burn or puncture skin, resulting in wounds that can cause blood poisoning, infection, disability, gangrene, amputation, or death. Therefore, the following instructions should be heeded at all times when inspecting or servicing the hydraulic components of the log splitter:

- Turn off the engine, move all control valve handles back and forth to relieve pressure, and disconnect hydraulic lines before changing or adjusting hydraulic system components such as hoses, tubing, fittings or other components.
- NEVER check for leaks with your hand. Leaks can be located by holding a piece of cardboard or wood (at least two feet long) with your hand at one end and passing the other end over the suspected area (wear eye protection). Look for discolouration of the cardboard or wood.
- If injured by escaping fluid, no matter how small the wound is, see a doctor at once. A typical injection injury may be a small puncture wound that does not look serious. However, severe infection or reaction can result if proper medical treatment is not administered immediately by a doctor who is familiar with injection injuries.

Problems

Problem	Solution
Cylinder rod will not move	A, D, E, H, J
Slow cylinder shaft speed when extending or retracting	A, B, C, H, I, K
Wood will not split or splits extremely slowly	A, B, C, F, I, K
Engine bogs down during splitting	G
Engine stalls under low load condition	D, E

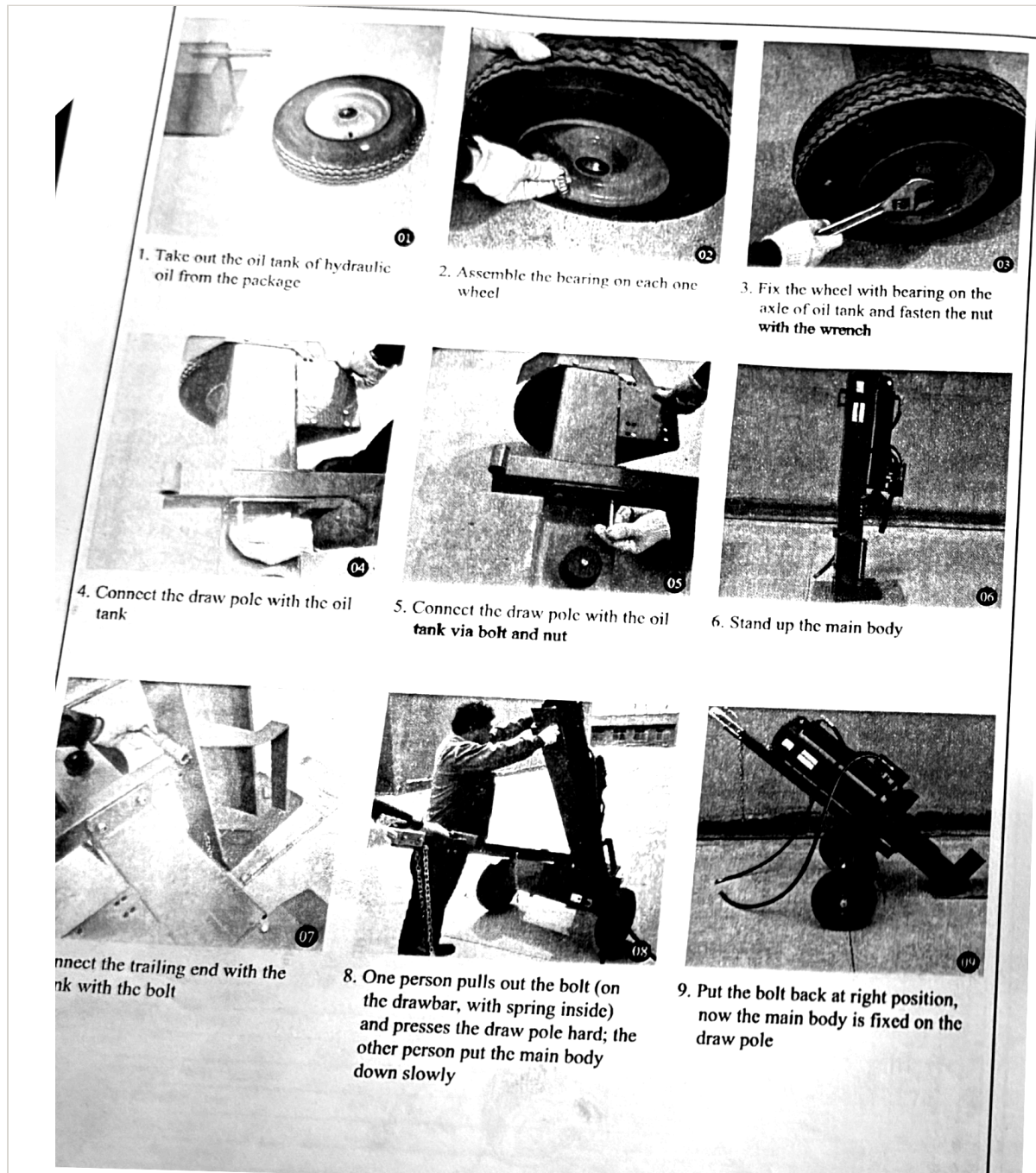
Causes & solutions

Cause	Solution
A — Insufficient oil to pump	Check oil level in reservoir
B — Air in oil	Check oil level in reservoir
C — Excessive pump inlet vacuum	Check pump inlet hose for blockage or kinks
D — Blocked hydraulic lines	Flush and clean the splitter hydraulic system
E — Blocked control valve	Flush and clean the splitter hydraulic system
F — Low control valve setting	Adjust control valve with a pressure gauge
G — High control valve setting	Adjust control valve with a pressure gauge
H — Damaged control valve	Return control valve for authorised repair
I — Internal control valve leak	Return control valve for authorised repair
J — Damaged cylinder piston	Return cylinder for authorised repair
K — Internally damaged cylinder	Return cylinder for authorised repair

SECTION 9

Assembly

Follow steps 1 to 18 in order. Two people are required for steps 8 and 9.



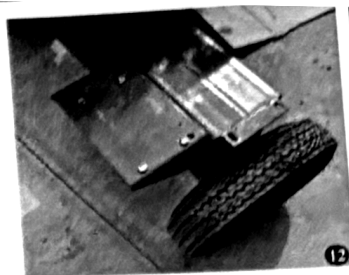
Assembly steps 1–9.



10. Assemble the rear jack



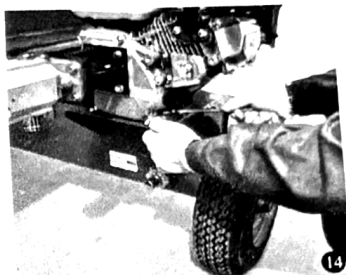
11. Keep the machine level by hand and assemble the front jack



12. Prepare the engine



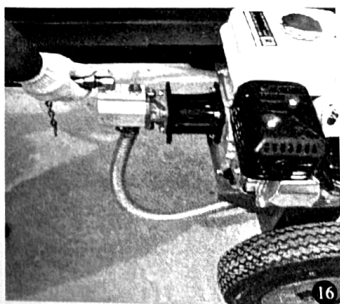
13. put the engine (with oil pump) at the correct position on the stand of machine.



14. Fasten the nut, 4 sets of nuts and bolts



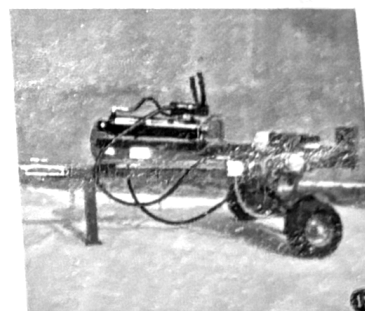
15. Connect the white color oil hose, between the oil pump (undersurface) and oil filter of oil tank and fasten the nut tightly



16. Connect the black color oil hose to the oil pump (bend connector of oil pump)



17. Connect the other black color oil hose to the oil tank



18. Complete machine photo

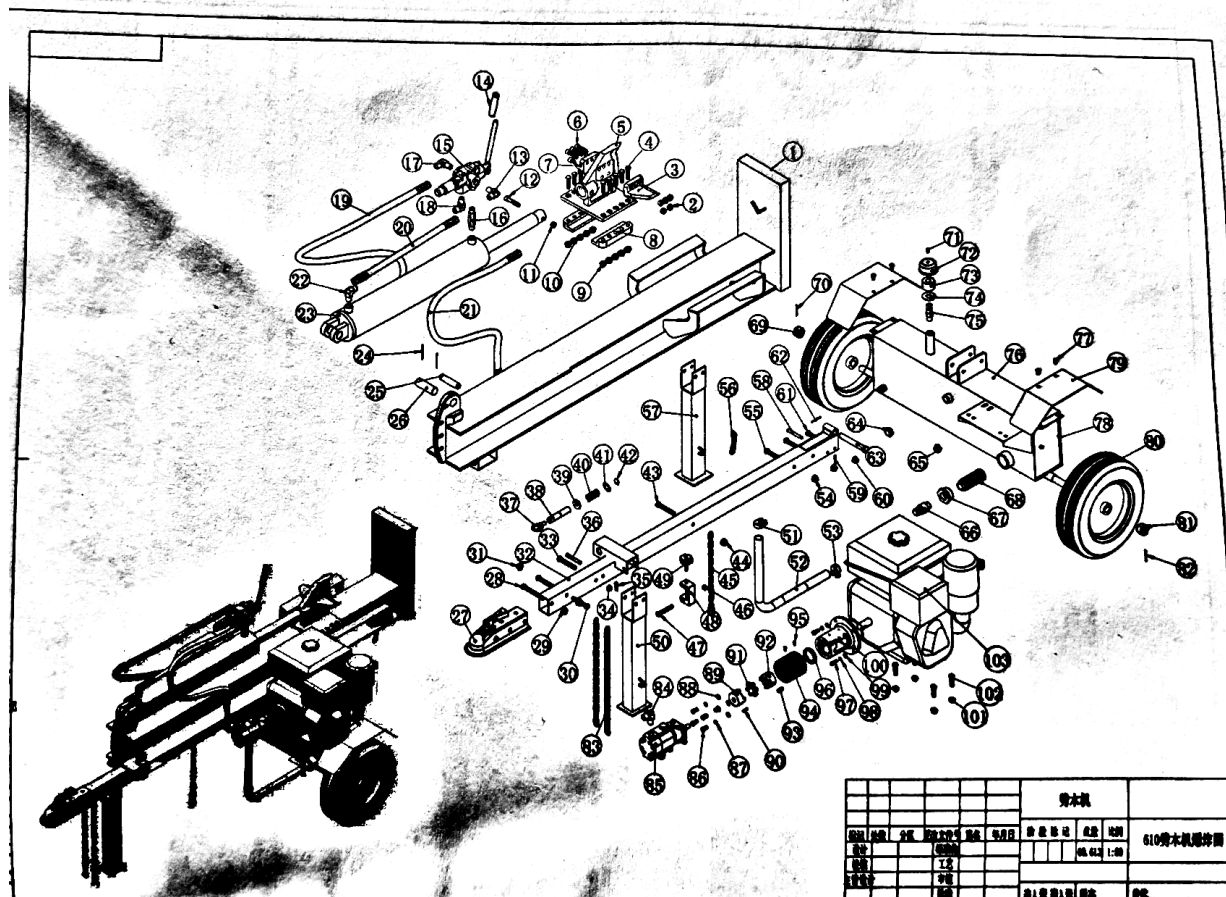
Please note, No.16, No.17, the oil hose should be connected by LEFT to LEFT, RIGHT to RIGHT, NOT CROSS.

Assembly steps 10-18.

Please note: at steps 16 and 17, the oil hoses must be connected LEFT to LEFT and RIGHT to RIGHT — do NOT cross them.

SECTION 10

Exploded parts diagram



Exploded view — quote the numbered item when ordering replacement parts.